

Submission deadline: May 31, 12:00 AM

Environment Selection

For this homework assignment, you must choose one environment from OpenAI Gym other than CartPole. Clearly state your chosen environment in your reports and investigate the observation space, actions, and the dynamics (if available).

Double Deep Q-Networks (DDQN)

This homework involves implementing Deep Q-Network (DQN) and its improvement, Double Deep Q-Network (DDQN). Standard DQNs suffer from **overestimation bias** due to their maximization step.

DDQN mitigates this by decoupling action selection from value estimation using two networks: an *online network* (θ_t) to choose the best action and a separate *target network* (θ_t^-) to evaluate that action's Q-value. This separation reduces overestimation and aims for more accurate Q-values, improved performance, and faster convergence. Below is the pseudocode algorithm for DDQN:

Algorithm 1 Double Deep Q-Network (DDQN)

- 1: Initialize replay memory D to capacity N
 - 2: Initialize action-value function Q with random weights θ (online network)
 - 3: Initialize target action-value function $\hat{Q}$ with weights $\theta^- = \theta$ (target network)
 - 4: **for** episode = 1, M **do**
 - 5: Initialize sequence s_1 and preprocessed sequenced $\phi_1 = \phi(s_1)$
 - 6: **for** $t = 1, T$ **do**
 - 7: With probability ϵ select a random action a_t
 - 8: Otherwise select $a_t = \arg \max_a Q(\phi(s_t), a; \theta)$
 - 9: Execute action a_t in emulator and observe reward r_t and next state s_{t+1}
 - 10: Set $s_{t+1} = s_t, a_t, x_{t+1}$ and preprocess $\phi_{t+1} = \phi(s_{t+1})$
 - 11: Store transition $(\phi_t, a_t, r_t, \phi_{t+1}, \text{done}_t)$ in D
 - 12: Sample random minibatch of transitions $(\phi_j, a_j, r_j, \phi_{j+1}, \text{done}_j)$ from D
 - 13: Let $a'_{\max} = \arg \max_{a'} Q(\phi_{j+1}, a'; \theta)$ (select action using online network)
 - 14: Set $y_j = \begin{cases} r_j & \text{if } \text{done}_j \\ r_j + \gamma \hat{Q}(\phi_{j+1}, a'_{\max}; \theta^-) & \text{otherwise} \end{cases}$ (evaluate action using target network)
 - 15: Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ with respect to the online network parameters θ
 - 16: Every C steps reset $\hat{Q} = Q$, i.e., set $\theta^- = \theta$
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Task

1. Choose your Gym environment.
2. Implement both the standard and Double Q-Network (DDQN) agents separately.
3. Train both agents in the same environment for the same number of episodes. Ensure other hyperparameters are kept as consistent as possible.
4. Plot the average reward per episode (or cumulative reward per episode) for both the standard DQN and DDQN agents.

Homework 5
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5. After training is complete, disable exploration and run the final trained agent for a fixed number of 10 episodes using the greedy policy derived from the trained agent. Calculate and record the average cumulative reward over these evaluation episodes. **Do not perform any training updates during this evaluation phase.**
6. Compare agents in terms of convergence speed, final performance, and training stability.
7. Investigate the impact of target network update frequency.

Extra Point

- Performing an ablation study on constant and decaying epsilon in the epsilon-greedy policy of the DDQN task has an extra point.
- Implementing the Boltzmann exploration strategy has an extra point.

Submission Guidelines

- Homeworks must be completed in **groups of four**.
- Each group must submit a single file containing a **PDF report**, including answers to discussion questions, visualizations, and code explanations, and a **Jupyter Notebook**, containing all code and results.